

EV Code Signing & Microsoft Driver Attestation Guide

RESEC SYSTEMS PRIVATE LIMITED · GlobalSign EV Code Signing (SafeNet token) · Prepared 27 Aug 2026

1. Environment summary

Item	Value
Certificate subject	RESEC SYSTEMS PRIVATE LIMITED
Issuer	GlobalSign GCC R45 EV CodeSigning CA 2020
Certificate SHA1 thumbprint	2CE7FCBEF04C279943F175A936E7588D298D254A
Expires	18 Aug 2027
Private key location	SafeNet USB token (SafeNet Authentication Client must be installed, token plugged in)
Cross certificate	C:\CodeSigning\r3csr45cross2020.crt
SignTool	C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe
Timestamp server	http://timestamp.globalsign.com/tsa/r6advanced1
Partner Center status	EV certificate registered (SignableFile.bin uploaded, status Active)

2. The standard signing command

This is the command that was verified working on this machine. Replace the last argument with the file you want to sign. Run it from a normal Command Prompt or PowerShell with the token plugged in; SafeNet will ask for the token password.

```
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" sign /v /fd SHA256 /ac "C:\CodeSigning\r3csr45cross2020.crt" /sha1 2CE7FCBEF04C279943F175A936E7588D298D254A /tr http://timestamp.globalsign.com/tsa/r6advanced1 /td SHA256 "C:\path\to\file.exe"
```

Switch	Meaning
/v	Verbose output (shows which certificate was picked).
/fd SHA256	File digest algorithm. Required by Microsoft; do not use SHA1.
/ac <crt>	Adds the GlobalSign cross certificate so the chain is embedded correctly.
/sha1 <thumbprint>	Selects the EV certificate explicitly. Always use this instead of /a ; /a picked a WDK test certificate on this machine.
/tr <url> /td SHA256	RFC 3161 timestamp. Keeps the signature valid after the certificate expires. Needs internet access.

Verify a signed file

```
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" verify /v /pa "C:\path\to\file.exe"
```

Expected result: `Successfully verified`, chain ending in *RESEC SYSTEMS PRIVATE LIMITED*, and a line `The signature is timestamped`.

3. Signing an EXE, DLL, MSI or other user-mode file

No Partner Center step is needed. Sign directly with the command in section 2, for example:

```
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" sign /v /fd SHA256 /ac  
"C:\CodeSigning\r3csr45cross2020.crt" /sha1 2CE7FCBEF04C279943F175A936E7588D298D254A /tr http://time  
stamp.globalsign.com/tsa/r6advanced1 /td SHA256 "C:\Build\MyApp.exe"
```

You can sign as many files as you like. Multiple files can be passed at the end of one command, or use a wildcard such as `"C:\Build\*.dll"`.

EV-signed executables receive immediate Microsoft SmartScreen reputation, so users will not see the "unknown publisher" warning.

4. Signing a kernel-mode driver (Microsoft attestation signing)

Windows 10/11 (64-bit) only loads kernel drivers that carry a Microsoft signature. Your EV certificate is used to sign the package you submit; Microsoft then returns the driver with its own signature. Repeat these steps for every driver package (and for every new build of it).

Step 1 – Prepare the driver package folder

Put all package files in one folder, e.g. `C:\CodeSigning\MyDriver\`:

- `MyDriver.inf` – the setup information file
- `MyDriver.sys` – the driver binary (64-bit)
- Any co-installer DLLs or other files referenced by the INF
- `MyDriver.cat` – optional; Microsoft creates its own catalog during attestation

Check the INF first with `"C:\Program Files (x86)\Windows Kits\10\Tools\10.0.22621.0\x86\infverif.exe" /v /w MyDriver.inf` (path may vary by SDK/WDK version). Partner Center rejects packages with INF errors.

Step 2 – Sign the driver binaries with the EV certificate

```
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" sign /v /fd SHA256 /ac
"C:\CodeSigning\r3csr45cross2020.crt" /sha1 2CE7FCBEF04C279943F175A936E7588D298D254A /tr http://time
stamp.globalsign.com/tsa/r6advanced1 /td SHA256 "C:\CodeSigning\MyDriver\MyDriver.sys"
```

Sign every `.sys` and `.dll` in the package the same way.

Step 3 – Build the CAB file

Create a text file `C:\CodeSigning\MyDriver.ddf` with this content (adjust file names and paths):

```
.OPTION EXPLICIT
.Set CabinetFileCountThreshold=0
.Set FolderFileCountThreshold=0
.Set FolderSizeThreshold=0
.Set MaxCabinetSize=0
.Set MaxDiskFileCount=0
.Set MaxDiskSize=0
.Set CompressionType=MSZIP
.Set Cabinet=on
.Set Compress=on
.Set CabinetNameTemplate=MyDriver.cab
.Set DestinationDir=MyDriver
.Set DiskDirectoryTemplate=C:\CodeSigning\out
C:\CodeSigning\MyDriver\MyDriver.inf
C:\CodeSigning\MyDriver\MyDriver.sys
```

Add one line per extra file (DLLs, etc.). Then build the cab:

```
makecab /f C:\CodeSigning\MyDriver.ddf
```

Output: `C:\CodeSigning\out\MyDriver.cab`.

Step 4 – Sign the CAB with the EV certificate (mandatory)

```
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" sign /v /fd SHA256 /ac  
"C:\CodeSigning\r3csr45cross2020.crt" /sha1 2CE7FCBEF04C279943F175A936E7588D298D254A /tr http://time  
stamp.globalsign.com/tsa/r6advanced1 /td SHA256 "C:\CodeSigning\out\MyDriver.cab"
```

Verify it:

```
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" verify /v /pa "C:\CodeSig  
ning\out\MyDriver.cab"
```

Partner Center rejects any CAB that is not signed with the registered EV certificate. The CAB signature must be made with the same certificate that was used for SignableFile.bin.

Step 5 – Submit to Partner Center

1. Sign in at <https://partner.microsoft.com/dashboard>.
2. Open **Hardware** → **Submit new hardware** (or **Drivers** → **Submit new hardware**).
3. Enter a product name (any label you like, e.g. *MyDriver 1.0.0*).
4. Upload `C:\CodeSigning\out\MyDriver.cab`. Upload it directly from the browser file picker; do not open, rename or copy it through email/chat first.
5. Under **Select the OS to sign for**, tick the Windows 10/11 versions and architectures you need. Attestation covers Windows 10 and Windows 11 client only.
6. Accept the attestation checkbox and click **Submit**.

Step 6 – Download the Microsoft-signed package

1. Wait for the submission status to become **Complete** (typically 10–30 minutes; refresh the page).
2. Click **More** → **Download signed files**. You get a ZIP containing your package plus a Microsoft-signed `.cat` file.
3. Distribute and install **that** package. Do not ship your locally signed copy; the Microsoft catalog is what allows the driver to load.

Install/test on a machine with `pnputil /add-driver MyDriver.inf /install` or `devcon`.

5. Notes and troubleshooting

Symptom	Cause / fix
<i>This file format cannot be signed because it is not recognized</i>	The input file is corrupt (binary was passed through a text conversion) or is not a PE/CAB/MSI. Re-download or rebuild the file. Quick check in PowerShell: <code>([IO.File]::ReadAllBytes('file') ? {\$_ -eq 0}).Count</code> must be greater than 0 for any PE file.
<i>No certificates were found that met all the given criteria</i>	Token not plugged in, SafeNet Authentication Client not running, or wrong thumbprint. Open SafeNet client and confirm the certificate is visible.

Wrong certificate selected	Never use <code>/a</code> on this machine; always use <code>/sha1 2CE7FCBEF04C279943F175A936E7588D298D254A</code> .
Timestamp error / <i>The specified timestamp server could not be reached</i>	Network/proxy issue. Retry, or temporarily use <code>/tr http://timestamp.digicert.com /td SHA256</code> .
Partner Center rejects the CAB	CAB not signed, signed with a different certificate, INF errors (run <code>infverif</code>), or missing 64-bit binaries.
Driver must run on Windows Server	Attestation is client-only. Windows Server requires HLK testing and a WHQL submission.
Token password prompt	SafeNet shows a dialog for every signing command. Do not let the token lock: repeated wrong passwords require the Administrator password/PUK to unlock.

6. Quick reference

```
# Sign (any file: exe, dll, msi, sys, cab)
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" sign /v /fd SHA256 /ac
"C:\CodeSigning\r3csr45cross2020.crt" /sha1 2CE7FCBEF04C279943F175A936E7588D298D254A /tr http://time
stamp.globalsign.com/tsa/r6advanced1 /td SHA256 "FILE"

# Verify
"C:\Program Files (x86)\Windows Kits\10\bin\10.0.22621.0\x64\signtool.exe" verify /v /pa "FILE"

# Build driver CAB
makecab /f C:\CodeSigning\MyDriver.ddf

# Driver workflow
sign .sys -> makecab -> sign .cab -> Partner Center upload -> download signed package -> i
nstall
```

Keep `C:\CodeSigning\r3csr45cross2020.crt` . The token must be present for every signing operation; the private key never leaves the token.